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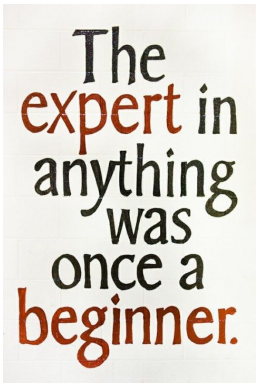
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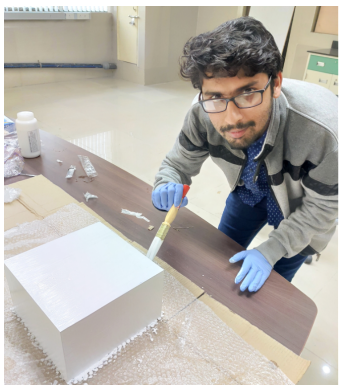
EIGHTH SEMESTER LAB PROJECT REPORT

BUILDING A SCINTILLATOR DETECTOR:

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Under The Supervision Of
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Contents

1 Abstract **3**

2 Introduction **4**

3 Objectives **5**

4 Theory **5**
- 4.1 Basic Working Of Scintillator Detector . . . 5
- 4.2 Organic Scintillators . . . 6
- 4.3 Scintillator Detector Components And Their Properties[4,5] . . . 7
  - 4.3.1 EJ-200 . . . 7
  - 4.3.2 EJ-510 Reflective Paint For Plastic Scintillators . . . 8
  - 4.3.3 Wrapping Materials . . . 8
  - 4.3.4 Silicone Rubber Optical Interface: EJ-560 . . . 9
  - 4.3.5 Photo-Multiplier Tube (PMT) . . . 9
- 4.4 Mean Life Time Of Stopped Cosmic Muons . . . 12
  - 4.4.1 Creation Of Cosmic Muons . . . 12
  - 4.4.2 Principle And Working Formula . . . 13
  - 4.4.3 Mean Range Of Muons . . . 13

5 Apparatus Required **15**

6 Experimental Setup **16**
- 6.1 Building A Scintillator Detector . . . 16
- 6.2 Stopped Cosmic Muon . . . 16

7 Observations **17**
- 7.1 Building A Plastic Scintillator Detector . . . 17
- 7.2 Mean Lifetime Of Stopped Cosmic Muon . . . 19

8 Calculations Data, Error And Graph Analysis **21**
- 8.1 Testing The Scintillator Detector . . . 21
- 8.2 Calculation Of Mean Life Time Of Stopped Cosmic Muon . . . 22
  - 8.2.1 Using The Original Setup . . . 22
  - 8.2.2 Using The Scintillator Detector We Built . . . 24
  - 8.2.3 Calculation Of Mean Value . . . 25
  - 8.2.4 Error Analysis . . . 26

9 Conclusion And Results **26**

10 Precautions And Sources Of Error **28**

11 Acknowledgements **28**

12 Additional Possible Future Experiments **29**

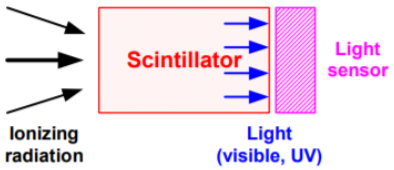
13 Appendix **30**
13.1 A:Example Root Code For Curve Fitting: 30
13.2 B:Data Sets For Stopped Cosmic Muon Mean Life Time 31

1 Abstract

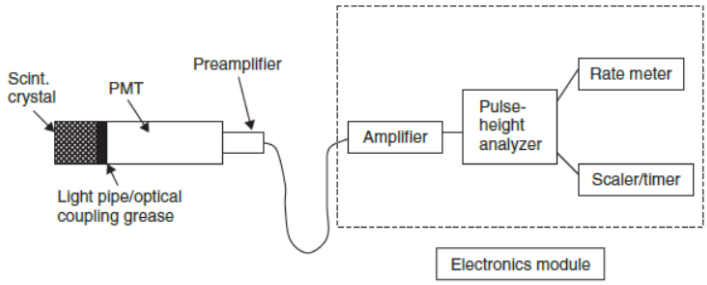
The idea of this experiment is to understand the working of a scintillator scintillator detector and then implement those working principles to make a plastic scintillation detector. A scintillation detector uses a scintillating material which produces a flash of light (photon) when struck by an incoming particle by absorbing their energy. This is coupled to a light sensor (PMT) was converts these light signals into electrical signals which can be analysed to gain information about the properties of the incoming particle. The scintillating medium in our case goes by the name EJ-200 (produced by Eljen technologies) which produces photons near the blue range of the visible spectrum. The scintillating medium is plastic having polyvinyl-toluene as the base which is doped with anthracene. The PMT model used is ET-9266 B which has a base radius of 52 mm and a quantum efficiency of around 30%. The PMT is protected with a plastic cylinder with inner and outer diameter being 5.875 *cm and* 7.610 cm respectively with an error of 0.0005 cm. To build the detector a variety of materials such as EJ-510 (reflective paint) and EJ-560 (Silicone Rubber Optical Surface) were used as their properties have a symbiotic relationship specific to the materials used in our detector. After the building phase is completed, the scintillator detector was brought to a testing phase where we simply checked if the detection is happening properly. The detector is directly connected to the oscilloscope and large and small peaks (corresponding to stopped cosmic muons and decay electrons) are observed in addition to constant background. This is exactly what was expected and our detector comfortably passed the testing phase.
Now, it would be a real shame if we did not perform any experiment with the detector we just built. In the next section, we then go on to perform the experiment of finding the mean life time of stopped cosmic muons.The fast response time, large available size of the crystal and high optical transmission makes our plastic scintillator detector a perfect fit to perform the stopped cosmic muon experiment. The experiment was performed both with the already available scintillator detector and our newly made detector. The values of the mean life time obtained from both the setups were quite close to each other and also the accepted value. The only difference we found while doing the experiment was that the original detector had 10 mV as its optimum discriminator settings while for our detector it was 15 mV. When we tried to work with the old settings on our detector we always got an unusually high reading near the 1274 channel of the MCA. We deduced that this must be due to some stray lights or noises. The experimental value of muon lifetime as calcuted in this experiment came out to be: $\tau_{exp} = 2.19056 \pm 0.068959\ \mu\ s$ which is well within the 1% error of the known value of $\tau_0 = 2.19703\ \mu\ s$.

2 Introduction

Scintillator detectors are one of the oldest types of radiation detector because measurements could be made with photographic film. Images could be collected or intensity measurements could be made. Measurements were also made with the human eye observing the brightness of frequency of flashes in the scintillator. Nowadays the light output is converted into voltage pulses that are processed in the same way as pulses from proportional counters. The whole point of scintillation detectors is that we want to produce a large light output in the visible range.
A scintillator is a material that illustrates the property of scintillation (Scintillation is a flash of light produced in a transparent material by the passage of a particle) using the property of luminescence (Luminescence is spontaneous emission of light by a substance not resulting from heat) when excited by ionizing radiation.
A scintillation detector or scintillation counter is obtained when a scintillator is coupled to an electronic light sensor such as a photomultiplier tube (PMT), photodiode, or silicon photomultiplier. PMT's absorb the light emitted by the scintillator and re-emit it in the form of electrons via the photoelectric effect. These are often known as primary photo-electrons. The subsequent multiplication of those electrons, generally using a dynode system (often called secondary photo-electrons) results in an electrical pulse which can then be analyzed and yield meaningful information about the particle that originally struck the scintillator. Vacuum photodiodes are similar but do not amplify the signal while silicon photodiodes, on the other hand, detect incoming photons by the excitation of charge carriers directly in the silicon. Silicon photomultipliers consist of an array of photodiodes which are reverse-biased with sufficient voltage to operate in avalanche mode, enabling each pixel of the array to be sensitive to single photons.



(a) Basic Principle



(b) Block Diagram

Figure 1: Scintillator Detector[1]

3 Objectives

I Understand The Working Of A Scintillator Detector[1]

II Building A Working Plastic Scintillator Detector For Stopped Cosmic Muons [1]

III Use the scintillator Detector we built to find the half life (Mean Life time) of the cosmic muons using the scintillator detector [2]

4 Theory

4.1 Basic Working Of Scintillator Detector

When an ionizing particle passes through the scintillator medium, the atoms get excited along a particular track. For a charged particle like an alpha particle, the track is the path of the charged particle itself. For high energetic rays like gamma rays, their energy can be converted into that of an energetic electron via the photo-electric effect, compton scattering or pair production.

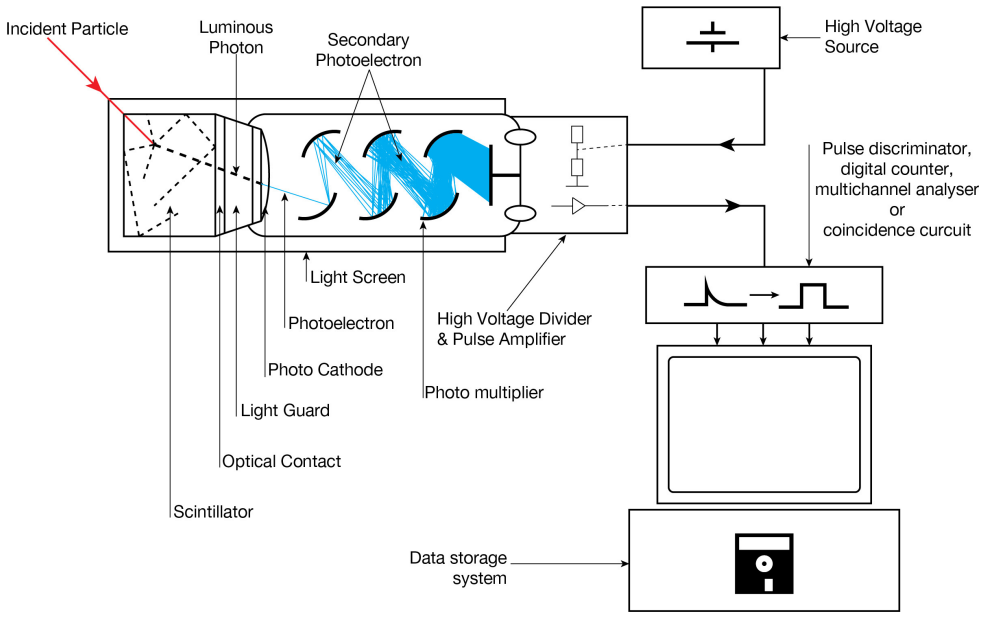


Figure 2: Operational Schematic Of A Scintillator Detector

The de-excitation of electrons in the scintillator produces photons generally near the blue region of the visible spectrum, The number of photons produced is directly proportional to the energy deposited by the ionising particle. These photons are directed towards a photo-cathode of the photo-multiplier tube to produce primary photo-electrons. Each arriving photon can at most produce one photo-electron

[1]Primary Objectives
[2]Secondary Objective

via the photo-electric effect. These primary photo-electrons are then focused and electrostatically accelerated by means of an electric potential to strike a dynode. The impact of a single electron on the dynode releases a number of secondary electrons which are in turn accelerated to strike the second dynode. Each subsequent dynode impact releases further electrons, and so there is a current amplifying effect at each dynode stage. Each stage is at a higher potential than the previous to provide the accelerating field.
The resultant output signal at the anode is a measurable pulse for each group of photons from an original ionizing event in the scintillator that arrived at the photo-cathode and carries information about the energy of the original incident radiation. When it is fed to a charge amplifier which integrates the energy information, an output pulse is obtained which is proportional to the energy of the particle exciting the scintillator. The number of such pulses per unit time also gives information about the intensity of the radiation.

4.2 Organic Scintillators

The fluorescence mechanism in organic materials arises from transitions in the energy levels of a single molecule and therefore the fluorescence can be observed independently of the physical state. For example anthracene is observed to fluoresce as a polycrystalline material, a vapor and even as a part of a mixed solution.

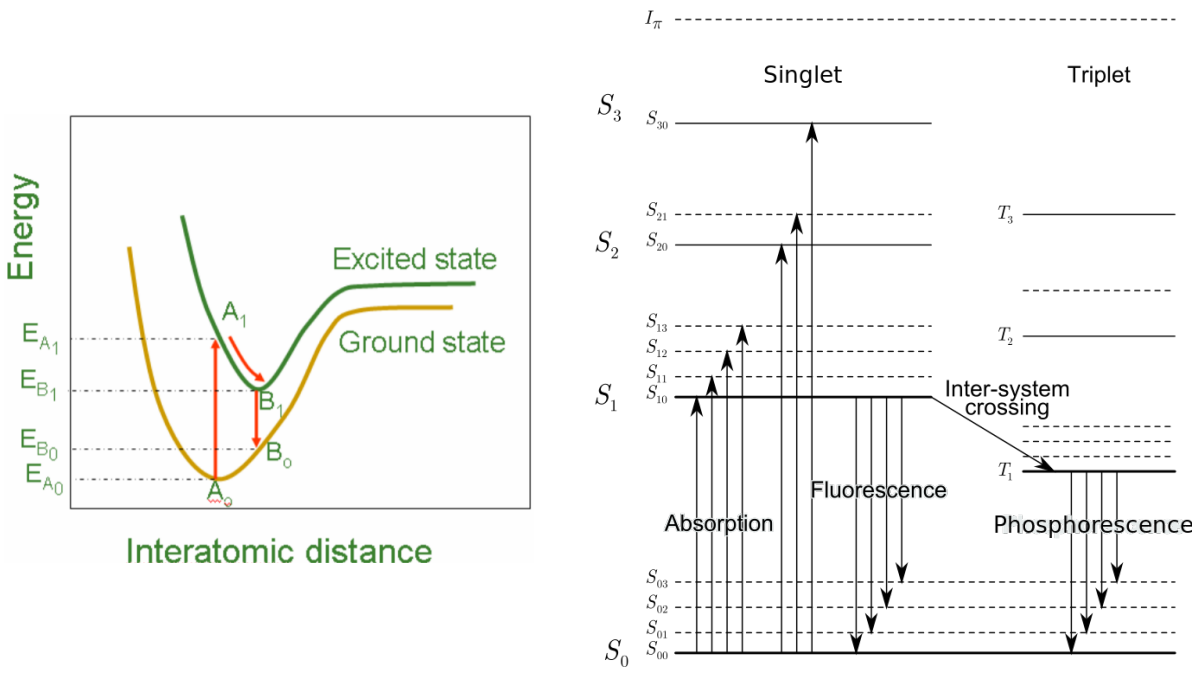


Figure 3: Energy Levels In An Organic Molecule [1]

The scintillation light in organic crystals is produced by the transitions of free valence electrons of the molecules. These electrons are part of the whole molecule and not of a single atom present in the molecule. Hence the electrons reside in the molecular orbitals given by MO theory. The ground state

is S_0 which is a singlet (spin 0) state above which are the excited singlet states, the first triplet (spin 1) state (T_1) and it's excited states. With each of these electron levels there exists fine structure levels corresponding to the molecular vibrational modes. For organic scintillators the spacing between S_0 and S1 is 3 to 4 eV, The spacing between adjacent electron reduces levels reduces as we take higher energy levels. Further, the typical spacing in vibrational structure is 0.15 eV. Note that the second label in the energy level figure denotes the vibrational level.

An incoming particle can excite either an electron level or a vibrational level. The singlet excitations from higher excited states (S_2 and above) quickly decay (<10 ps) to the S_{10} state via radiationless transfer (internal conversion). The S_{10} state then decays to the ground state S_0 (typically to one of the vibrational levels above S_0) by emitting a scintillation photon. This is known as fluorescence. Scintillation light, prompt fluorescence, is emitted in transitions between S_{10} and the ground state. The prompt fluorescence intensity at time t following excitation is described by decaying first order kinetics

$$I = I_0 e^{-\frac{t}{\tau}} \tag{4.2.1}$$

where τ is the fluorescence decay time for the S_{10} level to ground state.
Similarly when one of the triplet states gets excited, they quickly fall to the T_1 state via radiationless transfer.Since the process $T_1 \longrightarrow S_0$ is very improbable, the T_1 electron will decay by interacting with another T_1 electron via the following process

$$T_1 + T_1 \longrightarrow S_0 + S_1 + photon \tag{4.2.2}$$

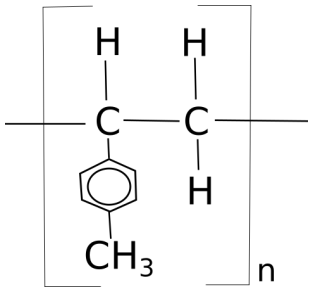
This leaves an electron in the S_1 state which can again decay to give a scintillation photon. Since $T_1 - T_1$ interaction takes time, the scintillation light coming from this process is delayed and hence often is known as delayed fluorescence. With very less probability the process (also delayed) $T_1 \longrightarrow S_0$ may also occur. The component of light coming from here is often termed as delayed phosphorescence. The energy level scheme explains why organic scintillators can be transparent to their own fluorescence emission. All fluorescence emissions (except $S_1 \longrightarrow S_0$) have a lower energy than the minimum required for absorption. There is little overlap between emission and absorption spectra, therefore the emitted light mostly passes straight on through the scintillation medium.
Organic scintillators can be dissolved in an organic solvent to form either a liquid or plastic scintillator. The scintillation process is the same as described for organic crystals. The difference arises in the mechanism of energy absorption where energy is first absorbed by the solvent and then passed onto the scintillation solute.

4.3 Scintillator Detector Components And Their Properties[4,5]

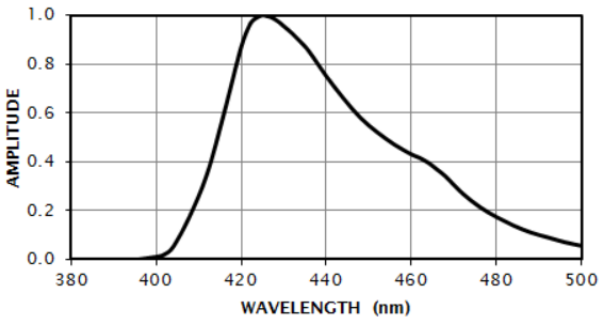
4.3.1 EJ-200

This is the scintillator material used in our experiment which produced photons when ionising radiation passes through it. The photons emitted were in the blue range of the visible spectrum. The material is a form of plastic with Polyvinyltoluene as the polymer base which is doped with anthracene. The decay time (τ in 4.2.1) for EJ-200 is 2.1 ns which makes it exceptionally good for creating fast timing scintillator counters. The material has a refractive index of 1.58 and a large light attenuation length

of 380cm. The emission spectra for our scintillator peaks at 425 nm (blue portion of the visible spectrum) which corresponds to the maxima of the emission spectra of the graph shown below. The scintillation efficiency is 10,000 photons/MeV which corresponds to a measly 2.9% of the total incident energy of the ionising radiation.This is generally the case and one of the disadvantages of using plastic scintillators.
EJ-200 combines the two important properties of long optical attenuation length and fast timing which make it particularly useful for time-of-flight systems using scintillators greater than one meter long. It is the detector of choice for many industrial applications, such as gauging and environmental protection, where high sensitivity and signal uniformity are critical operating requirements.
For sealed gas counters, an additional problem which arises is the large amount of quencher consumed in each detected event.



(a) Polymer Base- Polyvinyltoluene



(b) Emission Spectrum Of EJ-200

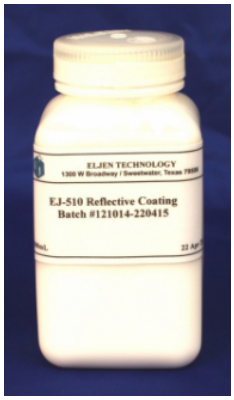
Figure 4: Properties of EJ-200 Scintillator

4.3.2 EJ-510 Reflective Paint For Plastic Scintillators

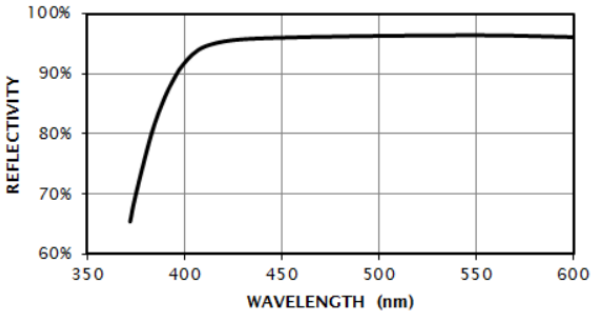
This is a bright white paint consisting of titanium dioxide pigment and a water soluble paint base selected for excellent resistance to yellowing and good adhesion. While primarily intended for coating of blue-emitting plastic scintillators, EJ-510 employs a blend of pigments selected also for enhanced reflectivity for longer wavelength scintillators with green emissions. This is a diffuse reflector for use on scintillators where the length is not greater than twice the width. It should not be used on long, narrow, optical elements. In addition to plastic scintillators, EJ-510 has been successfully applied to acrylic light guides and a variety of metals.
Maximum reflectivity is achieved by the application of three or four thin coats of EJ-510. After it has thoroughly dried, EJ-510 is not water soluble. However, it may be removed from a painted article by gentle rubbing with the aid of a mixture of water and isopropanol.
The reflectivity graph shown below makes this reflector paint a perfect match for making our scintillator detector.

4.3.3 Wrapping Materials

EJ-556 and EJ-558 are protective coverings used to protect plastic scintillator and acrylic light guide surfaces while in transport or during machining. EJ-556 is a paper backed covering and EJ-558 is a



(a) EJ-510



(b) EJ-510 Reflectivity Graph

Figure 5: Reflective Paint For Plastic Scintillators

low tack, transparent poly backed covering.
EJ-554 is a black vinyl adhesive tape used to make sealed and secure black vinyl wrappings for light-tight plastic scintillator detector assemblies.

4.3.4 Silicone Rubber Optical Interface: EJ-560

EJ-560 silicone rubber has been developed specifically for making optical joints between photo sensors and plastic scintillators. The rubber material is quite soft and flexible and can be made to conform to contoured surfaces. It is a fully-cured polymer designed so its surfaces are slightly sticky to the touch, and it can be deformed under mild pressure. The advantage of using silicone rubber is that it does not flow like a grease and will not extrude irreversibly out of its compression region, making it ideal for long-term coupling of photomultiplier tubes to scintillators.

EJ-560 is available in either pre-cut discs or sheet form. The sheet material can easily be cut to size with razor blades or scissors. EJ-560 rubber will easily adhere to any smooth surface and can also easily be peeled away without damaging either component. If the EJ-560 surfaces become covered with dust, their adhesive properties will be diminished. However, they can be cleaned with the aid of any lower alcohol, such as isopropyl alcohol and restored to their original condition.

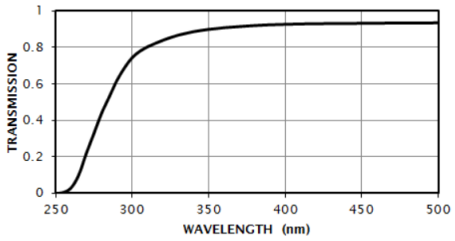
EJ-560 has a refractive index 1.43 which makes it an optically rarer medium with respect to EJ-200 (plastic scintillator). The transmission graph for EJ-560 tells that that the transmission factor reaches the maximum value for wavelengths above 400 nm. This makes it perfect for our scintillator since the peak of it's emission spectrum is 425 nm. Thus almost all photons that reaches the exit window passes through the PMT minimizing the energy losses.

4.3.5 Photo-Multiplier Tube (PMT)

In order to provide a useful measurement device, an electrical signal has to be formed from the scintillation light. A PMT is a device which is exactly meant to perform this task. PMT are great



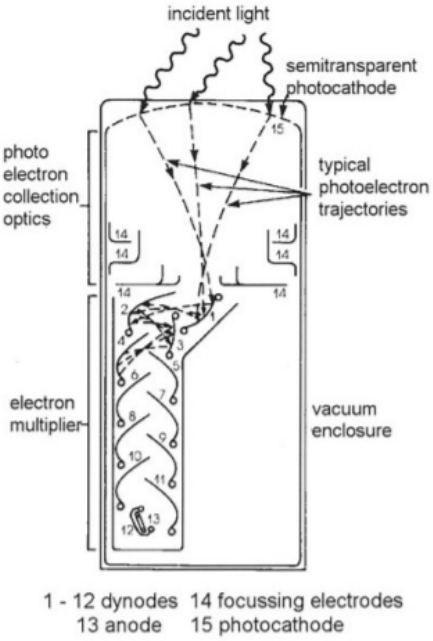
(a) EJ-560



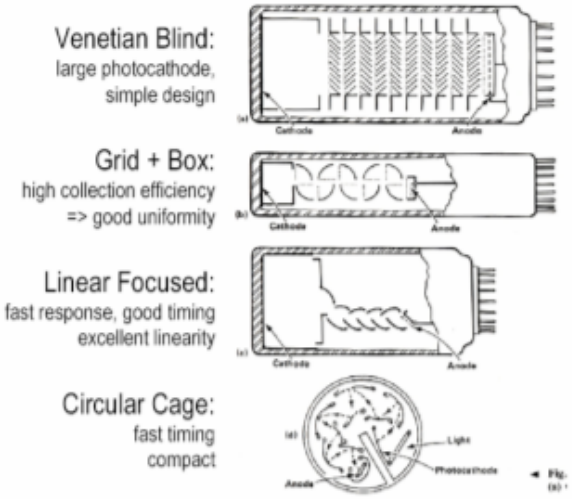
(b) EJ-560 Optical Transmission Graph (2mm thick)

Figure 6: Silicone Rubber Optical Interface

devices to convert light energy to electrical energy.A good PMT can produce a noticeable electrical signal from a few hundred visible photons.



(a) Basic Elements Of PMT



(b) Different Configurations Of PMT

Figure 7: Basic Properties Of Photo-Multiplier Tube

A PMT has two basic components:-

1. Photo-Cathode: Photomultiplier tubes are vacuum tubes in which the first major component is a photo cathode. A light photon may interact in the photo cathode to eject a low-energy electron into the vacuum via the photo-electric effect. This process can be divided into three main steps

- Absorption of the photon and transfer of energy to the electron in the photo cathode material

- Migration of the primary photo electron to the surface of the photo cathode
- Escape of primary photo-electron from the photo cathode surface.

The energy available to the scintillation photon to produce the photo-electron is around 3 eV. However, some of this energy is lost in electron electron collisions as the primary photo-electron migrates to the surface of the photo-cathode. To escape, the photo-electron must overcome the potential barrier at the vacuum surface interface called the work function. This poses limitations on the materials which can be used as a photo cathode. Typically metals have a work function of $3 \sim 4$ eV while semiconductors have a work function of $1.5 \sim 2$ eV.
The sensitivity of a photo cathode is usually quoted in terms of the quantum efficiency. The quantum efficiency of the photo cathode is defined as the probability for the conversion of light to an electrical signal and is defined as

$$Quantum\ Efficiency = \frac{No.\ of\ photo\ electrons\ emitted}{Number\ of\ Incident\ Photons} \tag{4.3.1}$$

The average quantum efficiency of a typical photo cathode over it's emission spectrum is 15-20 % and peaks around 25-30%.

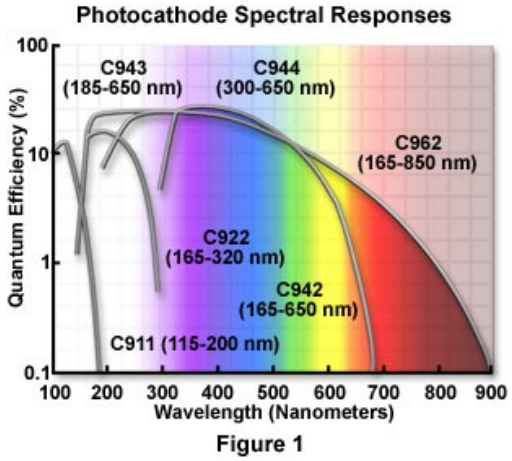


Figure 8: Photo Cathode Spectral Response

2. Electron Multiplication System Using A dynode Chain: Electrons are created in the photo cathode, and accelerated toward the first dynode, where secondary electrons are emitted. The secondary electrons are accelerated by the electric field toward a second dynode, where they create secondary electrons and this process is repeated multiple times.
Electrons are ejected from the photo cathode into the vacuum with an energy of ≈ 1 eV and are accelerated by a voltage of a few hundred volts toward an electrode. The accelerated electron has an energy of a few hundred electron volts upon arrival at the electrode; this deposition of kinetic energy can result in the re-emission of secondary photo-electrons. It typically takes 2-3 eV to excite an electron in the dynode, so 100 V can theoretically create ≈ 30 electrons. However, there is the same problem as in the photo cathode; electrons must be near the surface to be emitted, otherwise they lose

too much energy through collisions and are unable to overcome the potential barrier at the surface. Only a small proportion of the electrons excited up to the conduction band reaches the surface and creates secondary electrons. The overall multiplication factor for a single dynode is given by

$$\delta_{dmf} = \frac{No.\ of\ sceondary\ photo\ elctrons\ emitted}{No.\ of\ incident\ primary\ photo\ electrons} \tag{4.3.2}$$

PMT's use a multiple stage dynode multiplication system to create an overall gain of the order of 10^7 which is then converted to an electrical pulse and analysed using an external electronics system.
The datasheet of the PMT used in our experiment is provided in the following hyperlink: PMT ($https://drive.google.com/open?id=1pLAv6N_dj6cpXJ2YK67K7vDtipYrHxAX$) under the file name Data Sheet For PMT-9266B. (Note that the link to the google drive folder can only be accessed via a NISER mail id).

4.4 Mean Life Time Of Stopped Cosmic Muons

4.4.1 Creation Of Cosmic Muons

The cosmic radiation, which consists of high energy particles that are mostly protons, enters the earth's atmosphere and interacts with the atmospheric nuclei resulting in a shower of hadrons. Collisions of cosmic rays with atoms in the upper atmosphere produce mostly neutral and charged pions. The charged pions each decay, within less than 30 ns, into a muon and a muon neutrino (anti-neutrino). Another hadron called the Kaons are also produced during these collisions which also decay to muon and its' neutrino (anti-neutrino) in the order of 10^{-10} s about 64 % of the time. Both of the decay modes are weak interactions given by:

$$K^{\pm} \rightarrow \mu^{\pm} + \nu_{\mu}/\bar{\nu}_{\mu} \tag{4.4.1}$$
$$\pi^{\pm} \rightarrow \mu^{\pm} + \nu_{\mu}/\bar{\nu}_{\mu} \tag{4.4.2}$$

Muon itself is not a stable particle and hence tries to get converted into a more stable particle by interaction. To achieve this, there are two competing processes for muon:

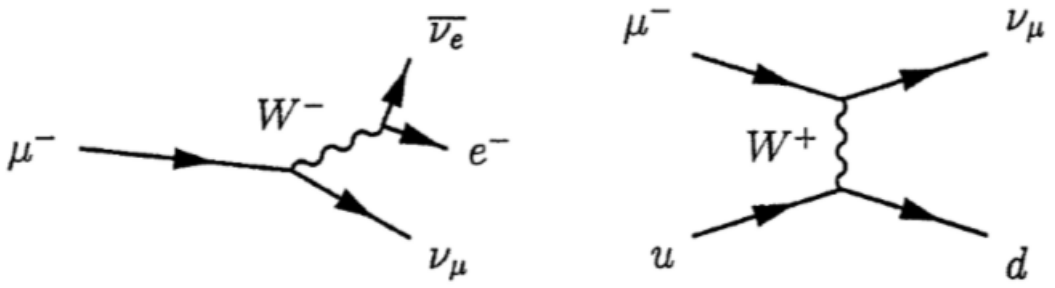
- Decay of muon: This interaction looks like:

$$\mu^{\pm} \rightarrow e^{\pm} + \nu_{\mu}/\bar{\nu}_{\mu} + \nu_{e}/\bar{\nu}_{e} \tag{4.4.3}$$

- Nuclear Capture of Muons: This interaction looks like:

$$\mu^{+} + n \rightarrow p + \bar{\nu}_{\mu} \tag{4.4.4}$$
$$\mu^{-} + p \rightarrow n + \nu_{\mu} \tag{4.4.5}$$

Due to positive charge on the nucleus, it is very unlikely that the positively charged muon will undergo nuclear capture as there will be a huge electrostatic potential barrier that will have to be overcomed. Hence, for μ^+ decay is the dominant process. However for μ^-, nuclear capture is quite a favourable process. This leads to a lower mean lifetime of μ^- when compared to μ^+.



(a) Decay of μ^-

(b) Nuclear Capture of μ^-

Figure 9: Feynman Diagrams For The Competing Processes Of Muon*[6]

4.4.2 Principle And Working Formula

Muons are assumed to decay spontaneously and therefore the probability of a muon decaying during any time interval dt is the same, and this probability is independent of the history of the muon. Muon decay just like any other decay in physics follows first order kinetics. This means that the number of decays in the time interval t-t+dt is proportional to the number of muons present at time t. Mathematically:

$$-\frac{dN}{dt} \propto N \tag{4.4.6}$$

The above equation can be integrated to give

$$N_t = N_0 e^{-\frac{t}{\tau_0}} \tag{4.4.7}$$

where τ_0 is the half-life (mean life in our case) of the muon. The latest measurements give an average value of $\tau_0 = 2.19703 \pm 0.00004\ \mu s$
A cosmic muon enters the scintillator and produces scintillations. This is referred to as the prompt signal. It eventually loses energy in the scintillator and decays into electron and neutrino when it stops. The electron produced by the decay again causes scintillations in the scintillator. This is referred to as the delayed signal. The time difference between the prompt and delayed signal is measured for a large number of events. The slope of the distribution of these time intervals gives the mean lifetime of muon. Uncorrelated events are also recorded since there may be accidental or chance coincidences with the prompt signal from background radiation, detector noise etc. These events are random in nature and therefore can be considered as a constant background.

4.4.3 Mean Range Of Muons

The mean lifetime of a stopped cosmic muon is about 2 μs. However they are produced thousands of meters away up in the atmosphere. Hence a question arises that how the muons are able to reach the sea-level and detected by our detectors. The answer to this is relativistic time dilation. To understand this we must differentiate between apparent life time and proper life time for a fast cosmic muon. The

lifetime of the muon is τ_0 in its rest system. However in the system where the muon appears to have moving with a velocity v appears to have a lifetime of $\tau_0^{'}$ which is related to the proper lifetime as:

$$\tau_0^{'} = \gamma\tau \qquad where\ \gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} \tag{4.4.8}$$

Hence, the mean range of the muons becomes:

$$R = v\tau_0^{'} = \frac{\tau_0}{m_\mu}p \tag{4.4.9}$$

where p is the relativistic momentum of the muon and m_μ is the rest mass of the muon. In the system of reference where the muon is moving, the apparent life of the muon is prolonged due to time dilation and therefore the muon can cover a range R, which could not be covered during the time τ_0. In the rest-system of the muon, however, the time available for the muon is only equal to τ_0. The range which was found to be equal to R in the system where the muon was moving appears to be contracted by a factor $\frac{1}{\gamma}$ in the rest-system. Therefore the muon has to live only for the time needed to pass the contracted range, $\frac{R}{\gamma}$.

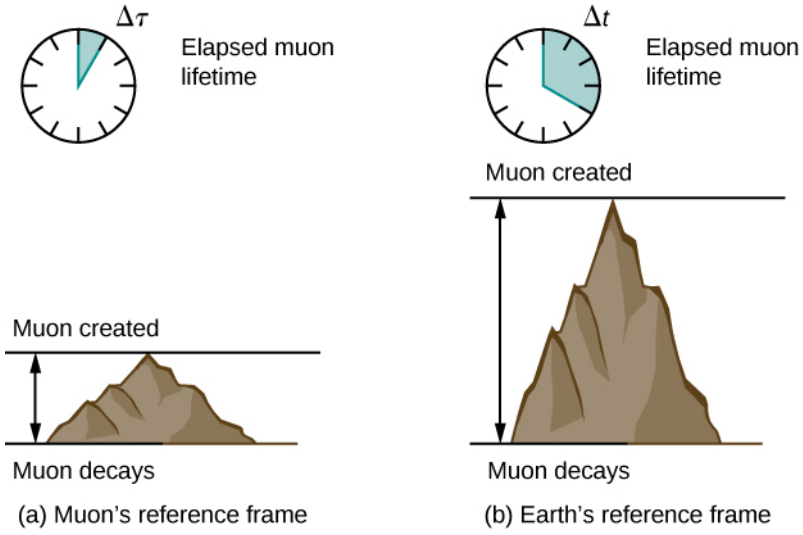


Figure 10: Relativistic Effects Allowing Muon To Reach The Earth

The Operational Manuals of the Nuclear Instrumentation Modules (NIM Setups) used in our experiment is provided in the following hyperlink: NIM Setups For Stopped Cosmic Muon Lifetime ($https://drive.google.com/open?id=12PK1QkHQsweGXmRgSxImYTqVVy492fpW$) under the appropriate file name. (Note that the link to the google drive folder can only be accessed via a NISER mail id).

5 Apparatus Required

- Scintillator Material (EJ-200 in our experiment)
- Photo-Multiplier Tube(PMT- ET-9266 B in our experiment)
- Reflective Paint For Plastic Scintillators (EJ 510 in our experiment)
- Silicone Rubber Optical Interface (EJ-560 in our experiment)
- Geometric Instruments
- Wrapping Materials (EJ 556 in our experiment)
- Sharp Blade
- BNC connectors
- Scintillator Detector (Plastic scintillator slab coupled to a Photo-Multiplier Tube)
- NIM crate - CAEN:Mod-N8303
- High Voltage Power Supply - CAEN:Mod-N1470
- LEMO coaxial cables
- BNC to LEMO converters
- Digital Storage Oscilloscope -Teledyne LeCroy: Mod- Wave Surfer 3034z
- Leading Edge Discriminator - CAEN:Mod-N841
- Multi Channel Analyzer - ORTEC:Mod-Aspec927
- Time to Amplitude converter - ORTEC:Mod-566
- Computer System with GECO 2020 software
- Pulse Generator

6 Experimental Setup

6.1 Building A Scintillator Detector



Figure 11: Scintillator Crystal

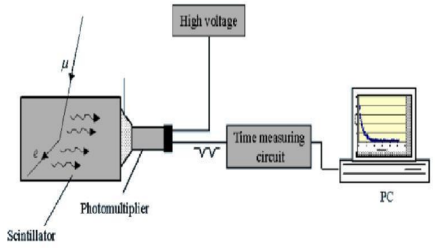
6.2 Stopped Cosmic Muon



(a) NIM Modules



(b) Scintillator Detector With Photo Multiplier Tube



(c) Block Diagram For The Setup

Figure 12: Laboratory Setup For Stopped Cosmic Muon

7 Observations

7.1 Building A Plastic Scintillator Detector

- Step 1: Take out the scintillating crystal (EJ 200) and put in on a soft smooth surface (We used bubble wrap over soft cardboard).
- Clean the surface of the crystal with cotton or tissue dipped in alcohol (iso-propanol was used). Try to not touch the surfaces of the crystal (especially the transparent one) as they might get scratched or you might leave oil/ fingerprint stains on the crystal. If you view the crystal along an edge you will be able to see faint bluish colored light coming out of the crystal.
- Step 2: Now take the specially designed reflective paint (EJ 510) and start coating the crystal with it. While painting, be sure to wear gloves as the paint is basically diluted titanium dioxide which is harmful for our body. While coating, try to maintain the same brush stroke on the surface of the detector. For best results, apply the coating 3-4 times. After each coating allow to paint to dry off for atleast 1 day. Do not start the next coating until the previous coating has completely dried.At this point all surfaces except the bottom one should have been coated.
- Step 3: Now apply black vinyl tap over the painted surface. This is done in order to stop any stray light or noise to go inside and produce scintillation in the detector (black body concept).
- Step 4: Now we need to make a connector/stand to join the PMT with the scintillator crystal. For this take precise and accurate measurements of the inner and outer radii of the PMT and the position and diameter of the screw holes on which the PMT is to be mounted. The connector is ring shaped (top view) and is made of acrylic. Flip the scintillator crystal carefully and clean the surface with alcohol gently. Connect the stand with the crystal using a special glue. In our experiment, the measurements came out to be:

  Outer Diameter Of The PMT: 7.610 cm
  Inner Diameter Of The PMT: 5.875 cm
  Diameter Of The Screw holes: 0.645 cm

- Step 5: Now take out the PMT and gently clean its surface with alcohol. Now take out EJ-560 (Silicone rubber optical interface), wash it with alcohol and place it in the hole of the acrylic connector directly touching the crystal. This acts as an exit window and EJ 500 acts as the surface (can be thought as some sort of light guide) guiding photons to the PMT.
- Step 6: Mount the PMT onto the scintillator crystal using EJ 500 and the acrylic stand. Once you are happy with the mounted position, tighten the screws to fix the assembly of PMT and scintillator.
- Step 7: As previously, apply 3-4 coatings of EJ-510 while maintaining all the precautions and wrap the remaining part of the crystal with black vinyl tape.
- Step 8: Check using the oscilloscope if detector is working properly. The oscilloscope should register some peaks if the detector is working. The scintillator detector is ready to find out the mean lifetime of stopped cosmic muons.



(a) Bare Scintillator Crystal



(b) After 3 Coatings Of EJ-510



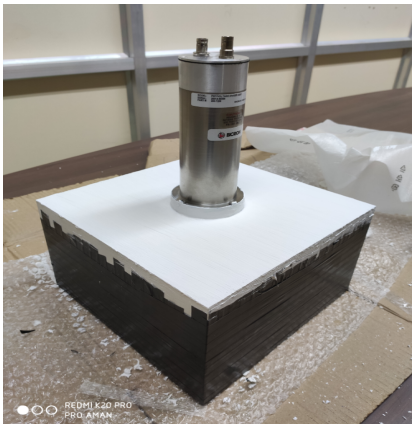
(c) Application Of Black Vinyl Tape



(d) Photo-Multiplier Tube (Inside View)



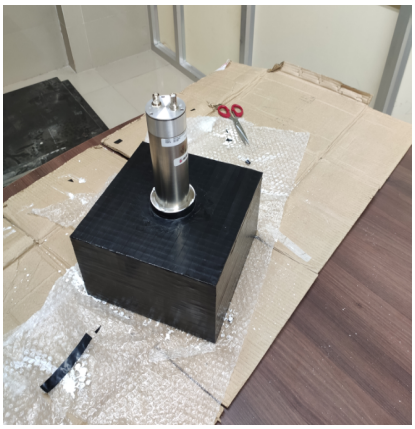
(e) Mounting The PMT Onto The Crystal



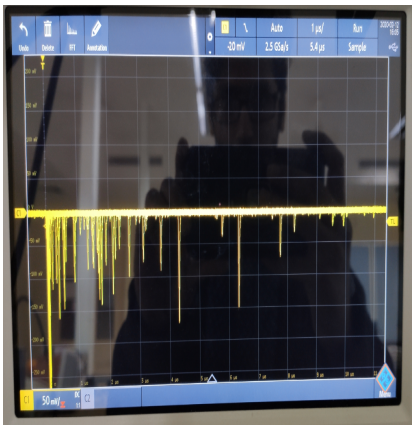
(f) After 3 coatings Of EJ-510 To The Bottom Surface



(g) Wrapping Black Vinyl Tape Onto The Remaining Surface



(h) Scintillator Detector Almost Ready For Detecting Cosmic Muons



(i) Oscilloscope Output When Detector Is Working Properly

Figure 13: Construction Of A Working Scintillator Detector

- The images in Fig 13 feature. 1- My lab partner Divyanshu Gupta in images (a) and (g) and 2- Scientific Officer Dr Varchaswi KS Kashyap in image (e).

7.2 Mean Lifetime Of Stopped Cosmic Muon

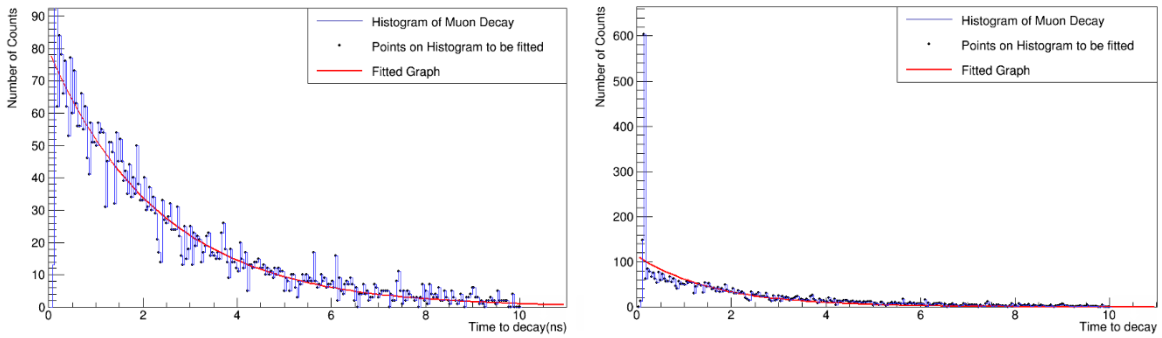
* The muon setup consists of three main NIM modules: Discriminator, Time To Amplitude Converter (TAC) and the Multi-Channel Analyser (MCA). In addition to this we have a desktop with appropriate analysing software along with a scintillator detector which is connected to the NIM modules using the LEMO cables.

* The mean life time of stopped cosmic muon is around 2 μs. They are generally formed about 20km above the sea level and the simple fact that they reach our detector is explained by time dilation which reinforces the theory of special relativity.

* Each scintillator detector has a different optimal discriminator setting. When we were using the original setup, the discriminator setting was 10 V. However when we used the scintillator detector which we built previously at 10 V, we got a lot of background noise. The optimum setting of the discriminator for our scintillator was at 15 V.

* The correct discriminator setting is quite intrinsic to the experiment as it basically sets the minimum energy cut-off. If the set value is too low then the assumed constant background ruins the decay trend and if is too high then we don't have enough events to get a good fitting with the theoretical decay curve. At optimal settings we get 2-4 events per minute.

* TAC is another module whose correct settings is intrinsic to the success of the experiment. TAC basically converts the decay time of the muon (recognized by the time difference between prompt signal (muon depositing the energy to scintillator) and delayed signal (muon decaying into electron which deposits energy to the scintillator)) into amplitude. Having prior knowledge that muon lifetime is somewhere in the μ s range we set the upper limit of TAC as 10 μ s.



(a) Optimal Settings

(b) Sub-Optimal Settings

Figure 14: Effect Of Discriminator Settings On The Final Output (Time is in ns range)

* The graph of counts vs time indeed looks like an exponential decay curve. This reinforces the fact that muon decay is independent of the history of muon and follows a first order kinetics.

* The count distribution mimics the exponential decay more accurately as the setup analyses higher number of events. From empirical observation, we found that our setup needed to run 2-3 days before we could effectively and accurately analyse the mean life time from the muon count distribution.

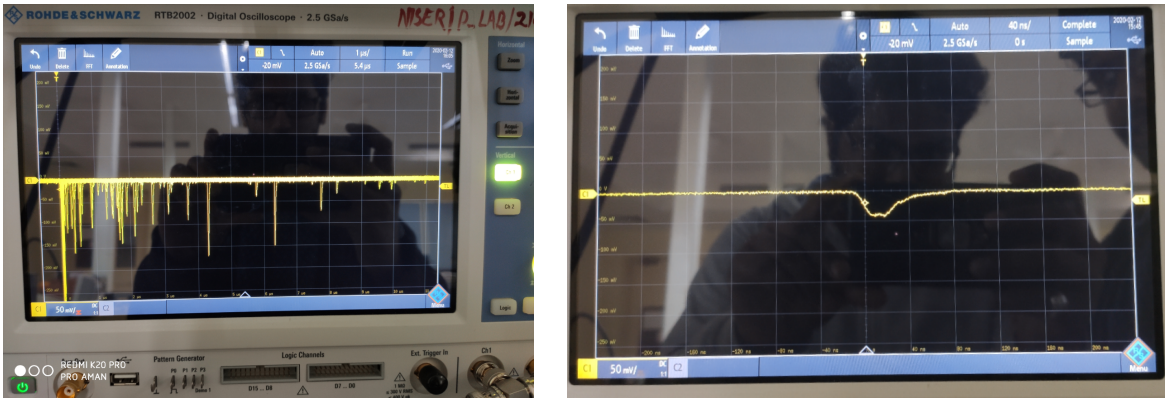
* The MCA must be calibrated properly using a known test function. In the TAC setup you can indeed set the maximum time (10 μs in our experiment) that will be recorded by the TAC. However this does not mean that the 10 μs pulse will correspond to 16384 (last bin) bin of the MCA. It is quite possible that this pulse is completely ignored by the MCA and the 16384 bin corresponds to 9 μs. Thus calibration with a known test function is necessary. For this a pulse generator can be used.

* After the calibration of the MCA, it was found that the last bin (16384) channel corresponded to 10.35 s and the first bin corresponded to 0.35 s. This is indeed a little different than the ideal values of 0 and 10 s at the 0 and 16384 bins/channels respectively.

8 Calculations Data, Error And Graph Analysis

8.1 Testing The Scintillator Detector

After completing the construction of the scintillator detector, it must be checked if it is detecting anything or not. This can done using a oscilloscope. Simply take the muon setup and instead of connecting the output of PMT to a time measuring circuit, connect it to the input of the oscilloscope while introducing an appropriate impedance into the ciruit (otherwise the oscilloscope might get damaged). The output is one of the following two graphs:



(a) Type 1 (b) Type 2

Figure 15: Possible Graph Outputs On Oscilloscope

In both of the graphs the x-axis is time while the y axis is the voltage. In type 1 graph, the oscilloscope is set to infinite persistence which means it records and stores all the outputs. There can be clearly seen two types of peaks on the oscilloscope. The larger peak corresponds to the muon striking the scintillator and the smaller peak corresponds to the decay of on to electron which then interacts with the scintillator. We know this as the voltage output on oscilloscope is proportional to the output of PMT which is in turn proportional to the energy of the incoming particle. Since muon must have a much larger energy than the electron it produces after decaying, hence the larger peaks must correspond to the muon interaction with the detector.
The type 2 graph is quite similar to type 1 graph . The only difference is that type-2 graph simply outputs the voltage corresponding to an interaction in the detector at the origin. This is useful if you want to observe the variation of the voltage peaks corresponding to the interaction in the detector. The small peak which can be seen in the above figure corresponds to the electron interaction (empirically determined from peak size with the same logic as before). Here is the video ($https://drive.google.com/open?id = 138DFq - tKMS0xP - Tnmo8V1BwZBzb_wF2S$) where type-2 graph was set to infinite persistence. Unfortunately, there is a lot of random background noise (as we have not connected the setup with a discriminator) due to which the graph becomes crowded quickly and nearly forms a uniform valley of points. However this indeed indicates that the scintillator detector is working properly as it is detecting any radiation which is producing the scintillating effect inside. As the noise is assumed to be uniform and random we indeed expect the the graph to form a uniform valley of points. (Note that the link to the google drive folder can only be accessed via a

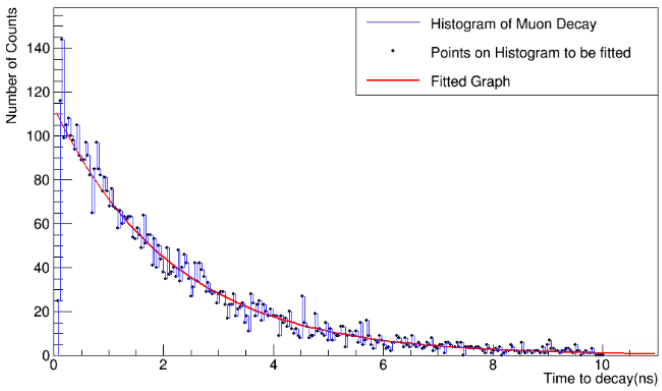
NISER mail id).
Hence the outputs of type 1 and type 2 graphs indicate that the scintillator detector we built is detecting properly (detecting what it should detect i.e. any particle which produces scintillation inside the detector) and the output is proportional to the energy of the particle producing the scintillation.

8.2 Calculation Of Mean Life Time Of Stopped Cosmic Muon

Now that the testing is done and completed, it is time to see our detector in action. Before testing our detector we perform the stopped cosmic muon experiment with the original setup present in our lab and then compare the results with our detector.

8.2.1 Using The Original Setup

The the original setup was used to take 3 different data sets. The data was already tabulated by the computer software which is provided in the appendix. The graphs/histograms are plotted using root and the same software was used to do least square regression to fit the obtained curve with the required exponential function. The fit curve was used to extract the necessary values of mean life time and the associated error in it.The fit function is given by 4.4.7 (N_0 is redefined to A in the fit function).



(a) Stopped Cosmic Muon Decay Distribution

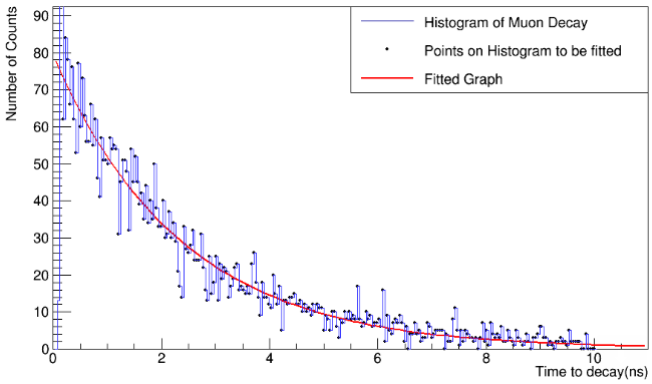
```
Minimizer is Minuit / Migrad
Chi2                      =      12693.1
NDf                       =          253
Edm                       = 1.46876e-010
NCalls                    =          120
A                         =      113.346   +/-   2.00031
Lifetime                  =      2.15395   +/-   0.0522601
```

(b) Values Of Appropriate Parameters And Errors In It

Figure 16: Data Set 1 For Stopped Cosmic Muon (Discriminator Value 10 mV) Using Original Setup

Data Set 1:
The value of stopped cosmic muon life time and the error after fitting the curve comes out to be:

$$\tau_{0(1,1)} = 2.15305 \pm 0.0522601 \quad \mu s \tag{8.2.1}$$



(a) Stopped Cosmic Muon Decay Distribution

```
**********************************
Minimizer is Minuit / Migrad
Chi2                      =      4650.66
NDf                       =          250
Edm                       = 4.55843e-010
NCalls                    =          121
A                         =      79.4587   +/-   1.30258
Lifetime                  =      2.35501   +/-   0.0513266
```

(b) Values Of Appropriate Parameters And Errors In It

Figure 17: Data Set 2 For Stopped Cosmic Muon (Discriminator Value 10 mV) Using Original Setup

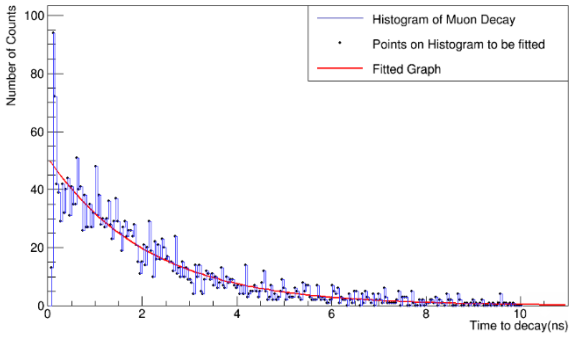
Data Set 2:
The value of stopped cosmic muon life time and the error after fitting the curve comes out to be:

$$\tau_{0(1,2)} = 2.25501 \pm 0.0513266 \quad \mu s \tag{8.2.2}$$

Data Set 3:
The value of stopped cosmic muon life time and the error after fitting the curve comes out to be:

$$\tau_{0(1,2)} = 2.10835 \pm 0.08609 \quad \mu s \tag{8.2.3}$$

As we can see from figures 15-17, the histogram indeed follows a good exponential decay. This tells us that the assumption that muon decay is independent of muon history as well as it follows a first order kinetics process was a reasonable assumption to make. The values obtained from each data set are quite close to each other and the known value of mean life.



(a) Stopped Cosmic Muon Decay Distribution

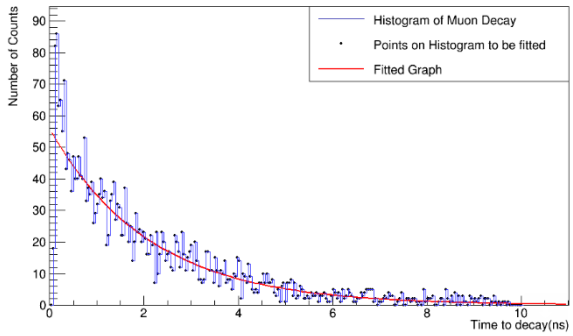
```
Minimizer is Minuit / Migrad
Chi2                      =      7132.38
NDf                       =          253
Edm                       = 4.03779e-007
NCalls                    =          117
A                         =      51.1871   +/-   1.51954
Lifetime                  =      2.10835   +/-   0.0860979
```

(b) Values Of Appropriate Parameters And Errors In It

Figure 18: Data Set 3 For Stopped Cosmic Muon (Discriminator Value 12 mV) Using Original Setup

8.2.2 Using The Scintillator Detector We Built

We used our scintillator detector to perform the stopped cosmic muon experiment and took 2 data sets with it. The graphs/histograms are plotted using root and the same software was used to do least square regression to fit the obtained curve with the required exponential function. The fit curve was used to extract the necessary values of mean life time and the associated error in it.The fit function is given by 4.4.7 (N_0 is redefined to A in the fit function).



(a) Stopped Cosmic Muon Decay Distribution

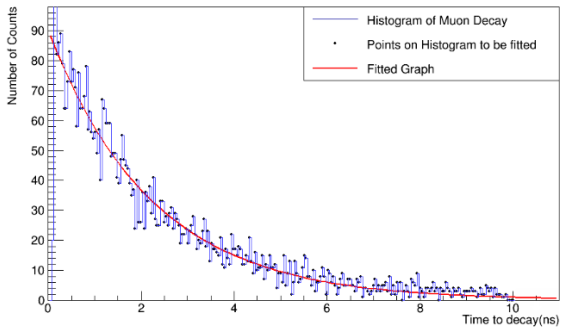
```
**********************************
Minimizer is Minuit / Migrad
Chi2                      =      9980.73
NDf                       =          254
Edm                       = 5.47494e-011
NCalls                    =          115
A                         =      56.0391   +/-   1.73989
Lifetime                  =      2.10615   +/-   0.0921463
```

(b) Values Of Appropriate Parameters And Errors In It

Figure 19: Data Set 1 For Stopped Cosmic Muon (Discriminator Value 13 mV) Using Our Detector

Data Set 1:
The value of stopped cosmic muon life time and the error after fitting the curve comes out to be:

$$\tau_{0(2,1)} = 2.10615 \pm 0.09214 \quad \mu s \tag{8.2.4}$$



(a) Stopped Cosmic Muon Decay Distribution

Minimizer is Minuit / Migrad				
Chi2	=	4397.49		
NDf	=	250		
Edm	=	4.937e-007		
NCalls	=	187		
A	=	90.3899	+/-	1.31065
Lifetime	=	2.23026	+/-	0.0429057

(b) Values Of Appropriate Parameters And Errors In It

Figure 20: Data Set 2 For Stopped Cosmic Muon (Discriminator Value 15 mV) Using Our Detector

Data Set 2:
The value of stopped cosmic muon life time and the error after fitting the curve comes out to be:

$$\tau_{0(2,2)} = 2.23026 \pm 0.04290 \quad \mu s \tag{8.2.5}$$

As we can see from Fig 18-19, the histogram again follows an exponential decay distribution, Further we can see from equations 8.2.1-8.2.5 that the mean life time values obtained from both the detectors are very close to each other and the known value of muon lifetime. Hence our detector passes with flying colors!!!

8.2.3 Calculation Of Mean Value

Now that we are armed with all the data. it's finally time to compute the experimental value of mean life time of stopped cosmic muons. This is done by simply averaging over all the values obtained through 5.2.1-5.2.5.

$$\tau_{exp} = \frac{\tau_{0(1,1)} + \tau_{0(1,2)} + \tau_{0(1,3)} + \tau_{0(2,1)} + \tau_{0(2,2)}}{5} \ \mu s$$

$$\tau_{exp} = 2.19056 \ \mu s \tag{8.2.6}$$

8.2.4 Error Analysis

- For Measurements In the PMT: All the measurements of the PMT (inner and outer diameter and diameter if hole) was done using a vernier caliper. The least count and hence the error in measurements is given by:

$$Least\ Count\ Of\ Vernier\ Caliper\ (LC) = 0.001\ cm \tag{8.2.7}$$

$$Error\ In\ Length\ Measurements\ (\Delta l) = \frac{LC}{2} = 0.0005\ cm \tag{8.2.8}$$

- For Muon Life Time: The error in life time can be defined in three in different ways:
  - Statistical Error: This is simply the standard deviation of the values used to calculate the muon lifetime.

$$(\Delta\tau_{exp})_{SE} = 0.068959\ \mu s \tag{8.2.9}$$

  - Propagational Error: As we are using uncertain measurements to calculate the value of muon life time, one could argue that those uncertainties propagate over the calculated value of muon life-time. This is not the right way to think about that formula however no one can stop you from calculating the error of propagation.

$$(\Delta\tau_{exp})_{PE} = \sqrt{(\Delta\tau_{0(1,1)})^2 + (\Delta\tau_{0(1,2)})^2 + (\Delta\tau_{0(1,3)})^2 + (\Delta\tau_{0(2,1)})^2 + (\Delta\tau_{0(2,2)})^2}$$

$$(\Delta\tau_{exp})_{PE} = 0.15201\ \mu s \tag{8.2.10}$$

  - Relative Percentage Error: One can also calculate the relative percentage as a means to show how close is our experimental value to the known and accepted value of that quantity. For muon lifetime the known value is $\tau_0 = 2.19703 \pm 0.00004\ \mu s$. Hence relative percentage error is:

$$(\Delta\tau_{exp})_{RE} = \left|\frac{\tau_0 - \tau_{exp}}{\tau_0}\right| * 100 = 0.29\% \tag{8.2.11}$$

The correct error to take is the statistical error because we are averaging over multiple independent values. Hence 8.2.9 gives the relevant error. The errors obtained in first two cases are quite low which suggests that we measurements are quite precise while the extremely low relative percentage error suggests that the value obtained is extremely accurate.

9 Conclusion And Results

- The principles behind the working of the scintillator detector were understood and implemented to build a working scintillator detector. As a bonus, the detector is then used to compute the mean life time of stopped cosmic muons.

- The scintillating crystal has the measurements 160mm x 250mm x 500mm (from shortest to longest side). They were given on the package in which the scintillating crystal arrived from Eljen Technologies. These dimensions give the crystal a large size which enhances it's ability to detect events useful to us.
- The scintillator crystal EJ-200 emits photon in the blue range of the visible spectrum. One can observe a faint blue light along the edge of the crystal at some particular viewing angle.
- The PMT used for our detector is of the E9266B series with a high quantum efficiency of 30% (from the available data sheet). The various measurements of the PMT are given as:

$$\begin{aligned} Outer\ Diameter\ Of\ The\ PMT: &\ 7.610 \pm 0.0005\ cm \\ Inner\ Diameter\ Of\ The\ PMT: &\ 5.875 \pm 0.0005\ cm \\ Diameter\ Of\ The\ Screw\ holes: &\ 0.645 \pm 0.0005\ cm \end{aligned}$$

- The scintillator detector is coated with special reflective paint EJ-510 to make no sure photon produced within the scintillator escapes without depositing the energy to the PMT. A black vinyl tape is used to make sure no stray light or noise goes into the detector. A silicone rubber optical interface EJ-560 along with an acrylic stand is used to connect the PMT to the scintillator. This produces a sturdy setup that can be used for a long time. The black vinyl tape provided to us was a simple electrical tape. It is advised to use an industrial grade black tape for a greater life of the detector.
- The built detector was first tested with oscilloscope to check if it was working properly. The analysis of section 8.1 tells that our detector passed with flying colors.
- The simple fact that the muons can be detected with our detector even though they are produced quite high in the atmosphere and have a measly life-time of 2.2 $\mu\ s$ is explained by the phenomenon of time dilation in special theory of relativity.
- The fast response time, large available size, ease of manufacturing and high optical transmission makes our plastic scintillator detector a perfect fit to perform the stopped cosmic muon experiment.
- The correct TAC settings, discriminator settings and correct calibration of the MCA is quite intrinsic to the success of the experiment.
- When we used our detector at 10 mV discriminator setting, we observed an unusually high peak near the starting end of the spectrum (channel 1274 had the peak value). This was corrected by covering the detector with a black cloth and changing the discriminator voltage. The noise was probably due to stray lights around the room.
- The optimum discriminator settings for the original detector is 10 mV and that for our scintillator detector is 15 mV.
- The value of mean life time of stopped cosmic muon from our experiment comes out to be:

$$\tau_{exp} = 2.19056 \pm 0.068959\ \mu\ s$$

  The value is quite close (within 1%) to the known value of $\tau_0 = 2.19703\ \mu\ s$.

10 Precautions And Sources Of Error

I Handle the scintillator detector carefully. Make sure to not scratch any surface of the detector. Do not forget to clean the surface with alcohol before painting it with the reflective paint.

II During each coating of the reflective paint, try to maintain the same brush stroke on the same surface. During alternative coatings change the brush stroke on the surfaces to get a better overall coating of reflective paint over the detector.

III After each coating allow the paint to dry for at least a day before starting the next coat. Do not apply the next coating unless the previous coating has completely dried.

IV While painting be sure to wear gloves. Try and not to waste the paint as it is quite expensive. Clean all the brushes and vessels after the coating is done.

V Paint the edges carefully. Try to maintain a uniform coating of the reflective paint along the edges.

VI Apply the black vinyl tape carefully on the edges. Make sure that none of the paint rips off during this process.

VII Measure the dimensions (inner and outer diameter) of the PMT as precisely and correctly as possible. Mark all the position of the screws and do not forget to take the diameter of the holes which will hold the screws.

VIII Handle the silicone rubber interface with care. Wash them with alcohol before using them.

IX The electronic modules and cables are expensive and not easily available. Rough handling is to be strictly avoided.

X Manuals of the modules should be read and thoroughly understood before connecting and performing the experiment.

XI HV applied to the detector must not be more than 1500V.

XII Set the discriminator at an appropriate setting else you will get a lot of unnecessary counts due to the background.

XIII Connect the circuit for stopped cosmic muons carefully. Make sure that the longer wire goes to start and the shorter to stop so that the circuit is resetted correctly each time.

11 Acknowledgements

The success and final outcome of this project required a lot of guidance and assistance from many people and I am extremely privileged to have got this all along the completion of my project. I owe my deep gratitude to our project supervisor Dr Subhankar Bedanta, who took keen interest on our project work and guided us all along, till the completion of our project work by providing all the necessary theoretical details and his own experimental expertise for developing a good system. I

would also like to thank Dr G Santosh Babu and Mr Ram Prasad and Dr Varchaswi KS Kashyap for their technical assistance in setting up the experiment and making sure that the instruments are in top working condition. I would like to extend my heartfelt esteems to all staff in laboratory for their timely support. Finally, I would like to express my sincere gratitude to my lab partner Mr Divyanshu Gupta for not only staying by my side and offering his experimental expertise in setting up the experiment but also providing the necessary emotional and mental support during frustrating moments of the experiment.

12 Additional Possible Future Experiments

- The students can be asked to make inorganic scintillator detectors (NaI doped with Tl is the most famous inorganic scintillating material). These detectors can be used in γ ray spectroscopy as inorganic scintillation crystals meet the requirements of stopping power and optical transparency. Their densities range from roughly 3 to 9 g/cm^3 making them very suitable to absorb penetrating radiation.
- Another interesting experiment using the plastic scintillation detector would be the detection of neutrons. Neutron being a neutral particle is difficult to detect and would make for an incredibly tough experiment if the students have to figure out how to do that from scratch. The detection of fast moving neutrons is based on the recoil photon produced in (n,p) reactions. Hence materials rich in hydrogen (plastic scintillators) are a match made in heaven for this detection.

References

[1] https://www.science.mcmaster.ca/radgrad/images/6R06CourseResources/4R6Notes4_ScintillationDetectors.pdf

[2] https://scionix.nl/wp-content/uploads/2017/12/SCIONIX-emission-spectra-scintillators-2.jpg

[3] https://en.wikipedia.org/wiki/Scintillation_counter

[4] https://eljentechnology.com/products/

[5] G.F. Knoll, Radiation Detection and Measurement - 3rd edition (Chapters 16 to 18), John Wiley Sons, 1999

[6] MUON LIFETIME MEASUREMENT,Serkant Ali Cetin ,Tuba Conka-Nurdan, Arif Mailov

13 Appendix

13.1 A:Example Root Code For Curve Fitting:

```
{
c1= new TCanvas("c1","Muon Lifetime",1000,600);
h1= new TH1F("h","Lifetime of Muon;Time(in microseconds);
Frequency",16384,0.035,10.045);
ifstream inp; double x;
Inp.open("Set 1.txt");
int i=0;
while (inp>>x)
{
h1->SetBinContent(i,x);
i++;
}

TH1F * hnew1 = dynamic_cast<TH1F *>(h1 ->Rebin(64,"hnew1"));
int d=h1->GetBinContent(112);
cout<<d;
hnew1->Draw();
hnew1->SetLineColor(kBlue-4);
hnew1->SetStats(0);
inp.close();
g1= new TGraph();
for(int k=5;k<=256;k++)
{
g1->SetPoint(k-5,hnew1->GetBinLowEdge(k),hnew1->GetBinContent(k));
}

fun1= new TF1("fun","[0]*exp(-x/[1])");
fun1->SetParNames("A","Lifetime");
fun1->SetParLimits(1,0.1,10);
g1->Fit(fun1);
g1->SetMarkerStyle(20);
g1->SetMarkerSize(0.6);
g1->SetMinimum(0);
g1->SetTitle("Discriminator=10mV,New Detector(Set 1);
Time to decay(ns);
Number of Counts");
g1->Draw("AP");
hnew1->Draw("SAME");

legend = new TLegend(0.5,0.7,0.90,0.90);
```

```
legend->AddEntry(h1,"Histogram of Muon Decay","L");
legend->AddEntry(g1,"Points on Histogram to be fitted","P");
legend->AddEntry(fun1,"Fitted Graph","L");
legend->Draw();
}
```

13.2 B:Data Sets For Stopped Cosmic Muon Mean Life Time

The number data points used for this experiment is insanely huge. Each data set has 16384 values. Each value corresponds to a bin/channel of the MCA which after calibration signifies some decay time t. The value corresponding to a particular bin/channel tells us the number of muons that decayed after time t. For eg: suppose the bin 1640 correspond to 1.2 μ s and it has a value of 80. This means that out of all the muons registered by the computer software, exactly 80 muons decayed after 1.2 μ s being stopped by the scintillator detector.

The data sheets that you will see are original computer files we obtained from the software. The software simply prints the value corresponding to each bin/channel of the MCA line by line. Hence the nth line of the notepad gives the value corresponding to the nth bin/channel of the MCA. Hence you will only see a list of values and nothing else. Please don't be alarmed as they are not supposed to be analysed analytically but with a software like root or python. Certainly only computers are efficient enough to handle, manage and play around with 16384 values in a matter of seconds.

Now that you know what to expect from our observational data sets, please click following hyperlink Data Sheets For Stopped Cosmic Muon to have a look at the original data sets [3]. (Note that the link to the google drive folder can only be accessed via a NISER mail id).

[3] ($https://drive.google.com/open?id=1ZaeCw0pjnS_rX656KKvMX5uyHi01YpiK$) is the html address of the google drive folder